

Active Recharting of Constraint Hypersurfaces

Orbit Quotients, Channel Accounts, and Physical Selection

A local-to-global synthesis with a Galois-spectral realization

William Lloyd

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Abstract

We assemble a local and a global branch into one theorem spine. Locally, a labelled three-coordinate constraint hypersurface is represented by its active output-chart jet orbit: a functional is invariant under active recharting exactly when it factors through the active orbit quotient. Curvature channel data forms a framed, defining-function-relative account whose canonical sum is an elementary shape-operator invariant. In the Kerr–Newman realization, positivity and interior regularity select a unique member of a stated inverse-channel metric family, and local metric germs classify its open boundary strata.

Globally, the four-charge seed-mass relation gives a finite signed-sheet cover. The axial k-ellipse pencil identifies the strict mass chamber with a definiteness chamber. An involutive parity block decomposition gives an exact singular-value factorization of the mass-square polynomial; every root is positive, every positive-imbalance sheet crosses once and transversely, and the physical root is the unique least root and first spectrahedral exit.

The unification claimed here is a theorem dependency and comparison, not an identification of group actions. The active chart group, radical sign group, axial deck involution, and global monodromy act on different carriers. Every point at which an identification, a generic-to-specialized arithmetic promotion, or a certificate-only calculation would be required is marked STOP-PROOF. The result is both a synthesis and an explicit proof programme.

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1 Introduction

1.1 Scope and terminology

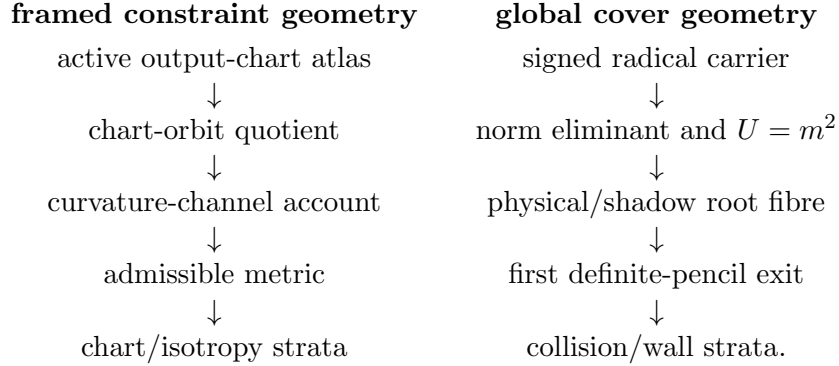
The abstract local datum in this paper is a constraint hypersurface equipped with a *labelled co-ordinate frame*. Its formal structure is the ability to use each coordinate as an output chart and compare the resulting finite jets under active recharting. In a directed application, the three coordinates may be interpreted as Directive, Substrate, and Product; we call that a *DBP realization*. Those semantics are not used in the abstract definitions or theorem names. This distinction is essential because the global Galois–spectral branch has no natural Directive–Substrate–Product interpretation.

1.2 The spine

The common logical order is

$\begin{aligned} &\text{carrier} \longrightarrow \text{orbit quotient} \longrightarrow \text{account fibre} \\ &\longrightarrow \text{admissible selection} \longrightarrow \text{typed strata} \longrightarrow \text{local normal form} \end{aligned}$

The local and global branches instantiate this order differently:



The central assertion is not that the columns are equal. Both exhibit a rigorous separation between invariant reduction and physical selection.

1.3 Status discipline

Established	A self-contained proof is included or reduces to a precise classical theorem whose hypotheses are checked.
Realization	Proved for a named family, not promoted to a universal statement for framed constraint hypersurfaces.
Bridge	An exact comparison or common architecture; no equivalence is asserted.
Conditional	Depends on a banked computation, certificate, or companion proof that must be re-verified before publication.

STOP-PROOF. *The word unified means an ordered theorem spine. Promoting the spine to an autonomous category requires native morphisms, equivalence, closure, and an independent third realization; none is supplied here.*

2 Framed constraint hypersurfaces and non-identification rules

2.1 The local chart datum

Let $F(x_1, x_2, x_3) = 0$ be a smooth local constraint relation. On

$$\mathcal{U}_1 = \{F = 0 : F_{x_1} F_{x_2} F_{x_3} \neq 0\},$$

the relation may be solved with each coordinate as output. Fix $r \geq 1$, write X_r for the regular finite-jet space, and $J_r(x)$ for one graph jet.

Definition 2.1 (active chart-jet carrier). *The group $G_{\text{act}} = S_3$ acts by re-solving the same relation in each output coordinate and transporting the jet by the finite inverse-function theorem. The carrier is*

$$\mathcal{O}_r(x) = G_{\text{act}} \cdot J_r(x).$$

Retaining coordinate values gives

$$\mathcal{O}_r^+(x) = G_{\text{act}} \cdot (x_1, x_2, x_3, J_r(x)).$$

2.2 Global cover data

For real parameters $(M, N_1, \dots, N_k)$ define

$$w_i(U) = \sqrt{U + N_i^2}, \quad s_\varepsilon(U) = \sum_i \varepsilon_i w_i(U), \quad \varepsilon \in \{\pm 1\}^k.$$

The signed norm is the polynomial

$$N_k(U) = \prod_{\varepsilon \in \{\pm 1\}^k} (4M - s_\varepsilon(U)),$$

and its oriented axial lift is $q_k(m) = N_k(m^2)$. We distinguish

$$H_{\text{sign}} = (C_2)^k, \quad G_{\text{deck}} = \langle \tau \rangle, \quad \tau(m) = -m,$$

from G_{act} and any global monodromy group; here C_2 is the cyclic group of order two.

Proposition 2.2 (non-identification guardrail). *The constructions do not provide canonical identifications*

$$S_3 \not\cong (C_2)^k, \quad X_r/S_3 \not\cong \text{Spec } \mathbb{Q}[U]/(N_k), \quad \Sigma_r \not\cong \text{Norm}.$$

Local chart recharting and global analytic-continuation monodromy also act on different sets.

Proof. The chart action permutes output charts of one local relation. Radical sign changes move signed sheets, and the deck involution changes an axial cover coordinate. No map intertwining these actions has been defined. $\square$

STOP-PROOF. *Any stronger claim requires an explicit carrier map, proof of equivariance, and comparison of stabilisers, branch loci, and quotients. A structural resemblance is not such a proof.*

3 The Active Recharting Orbit Theorem

Definition 3.1 (active-recharting invariant). *A function $\phi : X_r \rightarrow Y$ is an order- r active-recharting invariant if*

$$\phi(\mathcal{T}_\sigma \xi) = \phi(\xi)$$

for every $\sigma \in S_3$ wherever both sides are defined.

Theorem 3.2 (Active Recharting Orbit Theorem). *Let $\pi_r : X_r \rightarrow X_r/S_3$ be the set-orbit map. Then $\phi : X_r \rightarrow Y$ is an order- r active-recharting invariant if and only if there is a unique $\bar{\phi} : X_r/S_3 \rightarrow Y$ satisfying*

$$\phi = \bar{\phi} \circ \pi_r.$$

The same holds for the value-plus-jet carrier X_r^+ .

Proof. The active transition maps satisfy the S_3 relations on their common regular domain. A function is invariant exactly when it is constant on orbits, which is equivalent to unique factorization through the orbit set. $\square$

Proposition 3.3 (rational finite-jet action). *At orders one and two, active output-chart recharting is a rational S_3 action whose denominators are powers of the relevant first derivatives. At each fixed finite order the same conclusion follows recursively from the finite inverse-function theorem.*

STOP-PROOF. For an all- r headline, include the germ-level composition argument and a Faà-di-Bruno denominator induction. Otherwise restrict the published theorem to $r \leq 2$, where all formulas are explicit.

Remark 3.4 (mixed invariants). A derivative-only tuple is not automatically complete for invariants mixing values and jets. Such invariants belong on X_r^+ . The orbit theorem avoids claiming that a preferred finite kernel is complete without proving orbit separation.

4 Channel accounts and invariant reduction

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ define a regular hypersurface point and put

$$g = \nabla F, \quad H = \nabla^2 F, \quad q = g^\top g \neq 0.$$

In the distinguished coordinate frame, split

$$H = H_c + H_s, \quad H_s = \text{diag}(H), \quad H_c = H - H_s.$$

For $1 \leq r \leq n-1$ define

$$\hat{\mathcal{C}}_r(t, v) = (-1)^{r+1} \sum_{|I|=r+1} \det \begin{pmatrix} 0 & g_I^\top \\ g_I & (tH_c + vH_s)_I \end{pmatrix} = \sum_{p+s=r} t^p v^s \hat{\kappa}_{r;p,s}.$$

The letter v avoids collision with the mass-square coordinate $U = m^2$. Set

$$\kappa_{r;p,s} = \frac{\hat{\kappa}_{r;p,s}}{q^{(r+2)/2}}.$$

Theorem 4.1 (bordered-minor channel reduction). *Fix the oriented shape operator*

$$P = I - \frac{gg^\top}{q}, \quad S = -\frac{PHP}{\sqrt{q}}.$$

Then

$$\sigma_r(S) = \frac{\hat{\mathcal{C}}_r(1, 1)}{q^{(r+2)/2}} = \sum_{p+s=r} \kappa_{r;p,s}.$$

If $V_r = \mathbb{R}^{r+1}$ is the ambient coefficient space and

$$\Sigma_r : V_r \rightarrow \mathbb{R}, \quad (c_0, \dots, c_r) \mapsto \sum_j c_j,$$

then

$$0 \longrightarrow \ker \Sigma_r \longrightarrow V_r \xrightarrow{\Sigma_r} \mathbb{R} \longrightarrow 0$$

is exact. A geometrically realizable channel vector lies in the affine hyperplane over the reduced elementary extrinsic curvature σ_r .

Proof route. Rotate g to $\sqrt{q}e_n$. Only bordered minors containing the normal coordinate survive; each is $-q$ times the corresponding tangent principal minor. The sign and normalization give $\sigma_r(S)$. Replacing H by $tH_c + vH_s$ gives a homogeneous polynomial, and evaluation at $(1, 1)$ is the sum of its coefficients. Exactness is linear algebra. $\square$

STOP-PROOF. Write the rotated-frame bordered-minor calculation in full, including the global sign convention. Exact tests at finitely many dimensions confirm but do not replace the arbitrary- (n, r) proof.

4.1 Gauge transport

For $\tilde{F} = \mu F$ with $\mu(x) = 1$ and $a = \nabla \log \mu(x)$,

$$\tilde{g} = g, \quad \tilde{H} = H + ga^\top + ag^\top, \quad P\tilde{H}P = PHP.$$

Corollary 4.2 (gauge-channel transport). *The oriented shape operator and every σ_r are fixed, while*

$$C_r(\tilde{F}) - C_r(F) \in \ker \Sigma_r.$$

For $n = 3, r = 2$ this is

$$\delta\kappa_c + \delta\kappa_{\text{int}} + \delta\kappa_s = 0.$$

Remark 4.3. This proves that reachable gauge displacements lie in the zero-sum hyperplane. It does not prove that every vector in that hyperplane is geometrically reachable. If $\mu(x) < 0$, the chosen normal orientation reverses and odd σ_r change sign.

4.2 The Gaussian three-channel theorem

For a regular implicit surface in $\mathbb{R}^3$,

$$K_G = \kappa_c + \kappa_{\text{int}} + \kappa_s.$$

For a graph $z = h(x, y)$, with $Q_0 = 1 + h_x^2 + h_y^2$,

$$\kappa_c = -\frac{h_{xy}^2}{Q_0^2}, \quad \kappa_s = \frac{h_{xx}h_{yy}}{Q_0^2}, \quad \kappa_{\text{int}} = 0.$$

Thus the three active output-chart graph records lie on the one-dimensional slice

$$\kappa_{\text{int}} = 0, \quad \kappa_c + \kappa_s = K_G$$

inside the two-dimensional affine account plane over K_G .

STOP-PROOF. *The shape-operator spectrum is complete only for spectral scalar invariants under orthogonal conjugation. Do not call $\{\sigma_1, \dots, \sigma_{n-1}\}$ the complete set of all active-recharting invariants, which are governed by the larger chart-jet quotient.*

STOP-PROOF. *Channels are invariants of a framed defining-function jet, or canonical representatives after graph normalization. Fixing the coordinate frame does not fix defining-function gauge. Do not call raw channel values gauge-invariant.*

5 The quotient–selection separation theorem

Theorem 5.1 (two-realization quotient–selection separation). *The local chart and global cover realizations share the following exact separation.*

- (a) *In the local realization, the reduced invariant σ_r does not determine a channel representative whenever $\ker \Sigma_r \neq 0$. Graph normalization, defining-function gauge, or a metric construction supplies additional representative data.*

(b) *In the global realization, the polynomial $N_k(U)$ determines an unordered root multiset and does not by itself label one root as physical. On the strict chamber, an oriented definite pencil selects the physical root as its first positive boundary exit.*

(c) *Neither selection is part of the bare invariant quotient.*

Proof. Part (a) follows from the nontrivial affine fibre of Theorem 4.1. Polynomial coefficients are symmetric in the roots, so they do not carry a root label. The first-exit statement is proved in Theorem 7.3. Part (c) is the conjunction. $\square$

Remark 5.2. This theorem establishes a shared logical architecture. It does not provide a functor from channel fibres to root fibres.

6 Kerr–Newman inverse-channel metric selection

For the Kerr–Newman relation $M^2 = U(S, J, Q)$, each output coordinate i carries two mass–coordinate couplings and one coordinate–coordinate coupling. Denote the coordinate–coordinate weight by η :

$$g_F(\eta)_{ii} = q_i^2 \left(\Lambda_{i,\{M,j\}}^{-2} + \Lambda_{i,\{M,k\}}^{-2} + \eta \Lambda_{i,\{j,k\}}^{-2} \right).$$

Let

$$W_+ = \{S, J, Q > 0 : U_S > 0\}$$

be the outer physical wedge.

Theorem 6.1 (family-relative Kerr–Newman selection). *REALIZATION. Within the displayed weighted family:*

- (i) *each $\eta > 0$ produces an interior metric and curvature pole at a coordinate–coordinate coupling zero;*
- (ii) *each $\eta < 0$ loses positive definiteness near that zero;*
- (iii) *$\eta = 0$ is positive definite and interior-regular on W_+ , while retaining the desired mass–coordinate boundary divergences.*

Thus $g_F(0)$ is the unique admissible member of this family.

Proof route. The relevant coordinate–coordinate coupling has an interior zero. Its inverse square enters the metric with coefficient η . Positive η produces a metric pole and negative η forces the diagonal entry negative near the zero. At $\eta = 0$ that coupling is omitted. The remaining mass–coordinate couplings are finite and nonzero in W_+ and vanish only on distinguished boundary divisors in its closure. $\square$

STOP–PROOF. *Before Theorem 6.1 bears publication load, reproduce the exact six mass–coordinate factorizations, prove existence of the interior coordinate–coordinate zero, and archive a symbolic certificate that the $\eta > 0$ metric pole produces a nonremovable scalar-curvature pole.*

Corollary 6.2 (analytic corridor). *CONDITIONAL. The scalar curvature of $g_F(0)$ is analytic on W_+ , including the Davies surface, and its singular support in the closure is contained in*

$$\{T = 0\} \cup \{\Omega = 0\} \cup \{\Phi_e = 0\}.$$

STOP-PROOF. *To replace contained in by equals, prove nonvanishing of the leading residue on every open boundary stratum. The extremal edge depends on the full transverse-monotonicity and Sturm certificate.*

7 Selection realization II: algebraic normalization and reflection descent

This section separates generic function-field assertions from pointwise real assertions.

7.1 Generic algebraic theorem

Let

$$F_0 = \mathbb{Q}(N_1, \dots, N_k), \quad L = F_0(m), \quad w_i^2 = m^2 + N_i^2,$$

where the N_i are algebraically independent. Put

$$E = L(w_1, \dots, w_k), \quad s = \sum_i w_i, \quad F = F_0(M), \quad U = m^2.$$

Theorem 7.1 (generic normalization–reflection chain). *Assume the square classes of $m^2 + N_i^2$ are independent. Then:*

- (i) E/L is multiquadratic with group $H_{\text{sign}} = (C_2)^k$;
- (ii) the stabilizer of s in H_{sign} is trivial, hence $L(s) = E$;
- (iii) the orbit norm

$$\Phi_k(m, M) = \prod_{\varepsilon \in \{\pm 1\}^k} \left(4M - \sum_i \varepsilon_i w_i \right)$$

lies in $F_0[m, M]$, is even in m , and equals $N_k(U, M)$;

- (iv) the observable-field map induces

$$F[m]/(\Phi_k) \cong E;$$

- (v) $N_k(U)$ is irreducible over F and has degree

$$\delta_k = \frac{1}{2} \left(2^k - \mathbf{1}_{2|k} \binom{k}{k/2} \right);$$

- (vi) the involution

$$\tau(m) = -m, \quad \tau(w_i) = w_i$$

has fixed field $E^\tau = F(U)$, and every w_i descends to $F(U)$.

Proof route. Square-class independence gives the multiquadratic extension. A nontrivial sign change changes s , so the primitive-element stabilizer is trivial. The orbit product is invariant under all radical signs and even under $m \mapsto -m$. Irreducibility and the field identification follow from the incidence map and localization over $F_0[M]$. The fixed-field statement uses the degree-two reflection quotient. $\square$

STOP–PROOF. *The companion draft says that birational affine curves have isomorphic normalizations, which is not sufficient. If affine normalization is claimed here, prove that the incidence curve is normal, the map to the axial curve is finite and birational, and therefore has the normalization universal property. The field isomorphism itself does not require that stronger geometric sentence.*

7.2 The pointwise real theorem

Let X_i, Z_i be Pauli matrices acting in the i th tensor slot and put

$$L_0 = 4MI + \sum_i N_i X_i, \quad Z = \sum_i Z_i, \quad L(m) = L_0 - mZ.$$

This is the sign-conjugate form of the Nie–Parrilo–Sturmfels pencil, with

$$\det L(m) = N_k(m^2).$$

Define

$$\mathcal{C} = \left\{ (M, N) : 4M > \sum_i |N_i| \right\}.$$

Lemma 7.2 (chamber equals definiteness).

$$L_0 \succ 0 \iff (M, N) \in \mathcal{C}.$$

Proof. The eigenvalues of L_0 are $4M + \sum_i \varepsilon_i N_i$; the least is $4M - \sum_i |N_i|$. □

Let $X_{\text{all}} = \sigma_x^{\otimes k}$. It commutes with L_0 and anticommutes with Z . In its two eigenspaces,

$$L_0 = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & C \\ C^\top & 0 \end{pmatrix}, \quad A, B \succ 0.$$

Set $T = A^{-1/2}CB^{-1/2}$.

Theorem 7.3 (spectral chamber and physical selection). ESTABLISHED. *Let*

$$D_k = \begin{cases} 2^k, & k \text{ odd}, \\ 2^k - \binom{k}{k/2}, & k \text{ even}, \end{cases} \quad \delta_k = D_k/2.$$

For every real $(M, N) \in \mathcal{C}$, including zero or repeated charges:

(i) $\text{rank } C = \delta_k$ and

$$N_k(U) = \det(L_0) \prod_{j=1}^{\delta_k} (1 - U\sigma_j(T)^2);$$

(ii) every root of N_k is real and strictly positive;

(iii) each sign vector with positive imbalance $a_\varepsilon = \sum_i \varepsilon_i > 0$ crosses the level $4M$ exactly once and transversely, while zero- and negative-imbalance sheets do not cross;

(iv) the all-plus root U_{phys} is the unique least root;

(v)

$$U_{\text{phys}} = \lambda_{\max} \left(L_0^{-1/2} Z L_0^{-1/2} \right)^{-2} = \left[\max_{v \neq 0} \frac{v^\top Z v}{v^\top L_0 v} \right]^{-2};$$

(vi) if $N_k(U) = \sum_{r=0}^{\delta_k} c_r U^r$, then $(-1)^r c_r > 0$ for every r .

Proof. Parity makes the normalized symmetric pencil direction off-diagonal. Its nonzero eigenvalues are the pairs $\pm \sigma_j(T)$, yielding the factorization and positive roots.

Every signed sheet begins below $4M$ at $U = 0$. A positive-imbalance sheet tends to $+\infty$, so crosses at least once. The number of such sheets is $\delta_k = \deg N_k$; their crossings already saturate total degree counted with multiplicity. Hence each is unique and transverse and there are no further crossings.

At a shadow crossing,

$$\sum_i w_i(U) = 4M + 2 \sum_{\varepsilon_i = -1} w_i(U) > 4M = \sum_i w_i(U_{\text{phys}}).$$

Strict monotonicity of the all-plus sum gives $U > U_{\text{phys}}$. The generalized-eigenvalue formula follows, and coefficient alternation is the expansion of the positive singular-value product. $\square$

Corollary 7.4 (four-channel global simplicity). *For $k = 4$, if $N_i^2 \neq N_j^2$ for $i \neq j$, then all five roots are simple at every strict-chamber point. The physical root is first, and larger $|N_i|$ gives the later associated single-minus shadow root.*

Proof. The positive-imbalance sheets are the all-plus sheet and four single-minus sheets. Each crossing is transverse. Two single-minus sheets collide exactly when $w_i = w_j$, equivalently $N_i^2 = N_j^2$. $\square$

Corollary 7.5 (physical analytic section). *The map $(M, N) \mapsto U_{\text{phys}}(M, N)$ is real-analytic on the strict chamber and is fixed by continuation around every loop contained in that chamber.*

Proof. The physical factor has nonzero U -derivative everywhere in the chamber, so the implicit-function theorem gives local analyticity. Global uniqueness glues the local sections and prevents nontrivial continuation. $\square$

STOP-PROOF. *State explicitly that this is the sign-conjugate NPS pencil. Do not call the sign convention character-for-character identical to the source.*

STOP-PROOF. *The critical-sheet discriminant mechanism is absent for strict-chamber axial crossings, but it remains a genuine mechanism off chamber and in the complex/global discriminant. Do not erase it globally.*

8 Four-charge arithmetic consequences

Let $F = \mathbb{Q}(M, N_1, N_2, N_3, N_4)$. The generic $N_4(U)$ has degree five.

Theorem 8.1 (S_5 obstruction). **CONDITIONAL.** *Assuming the exact Dedekind certificates described in the companion calculation, N_4 is irreducible over F with generic Galois group S_5 . Consequently the generic seed mass-square is not expressible by radicals over the observable field.*

Certificate route. At one rational specialization, exact irreducibility makes the specialized group transitive, while a good-prime factorization of cycle type $(2, 3)$ forces that transitive quintic group to be S_5 . The separable specialized group embeds into the generic group, which is a subgroup of S_5 . $\square$

STOP-PROOF. *Include the specialized quintic, irreducibility certificate, good-prime verification, and modular factorization in the paper or a permanent machine-readable supplement. A prose report that a run occurred is not an auditable certificate.*

Corollary 8.2 (totally real arithmetic, correctly scoped). *Every strict-chamber specialization has five positive real roots counted with multiplicity. Any rational or algebraic chamber specialization that is separately certified irreducible with Galois group S_5 defines a totally real S_5 quintic field.*

Remark 8.3 (specialization guardrail). Generic S_5 does not persist at every chamber point. At $(M, N_1, N_2, N_3, N_4) = (2, 1, 1, 1, 1)$,

$$N_4(U) = 2^{30}(3 - U)(15 - U)^4.$$

The physical root is simple while four shadow sheets collide.

Proposition 8.4 (ordered horizon reconstruction). **CONDITIONAL.** *Under the generic descent and square-class hypotheses, signed radicals lie in $F(U)$, the oriented axis field is obtained by adjoining m , and the ordered horizon field is generated by either normalized horizon entropy. The rational entropy product reconstructs its conjugate.*

STOP-PROOF. *Keep unconditional field identification separate from the proposed degree-20 count. The latter still requires an independent square-class gate and a good-specialization argument.*

9 Involution, parity, and local curvature valuations

The cover theorem uses an involution commuting with the base matrix and anticommuting with the pencil direction. The inverse-channel boundary theorem uses a coordinate reflection fixing a metric germ. In both cases odd data vanishes, but the actions and outputs remain distinct.

Lemma 9.1 (fixed-germ parity). *If a scalar invariant $R(x, y)$ is constructed naturally from a metric germ whose components are even in x , then $R(-x, y) = R(x, y)$. Its Laurent expansion contains no odd powers of x .*

Proof. The reflection is an isometry of the germ. Naturality of scalar curvature gives the result. $\square$

Proposition 9.2 (generic quadratic collapse). *Let a diagonal metric satisfy*

$$g_{xx} = Ax^2 + O(x^3), \quad g_{\alpha\alpha} = P_{\alpha 0} + P_{\alpha 1}x + O(x^2), \quad AP_{\alpha 0} \neq 0.$$

Then

$$R = \frac{1}{A} \sum_{\alpha=1}^m \frac{P_{\alpha 1}}{P_{\alpha 0}} x^{-3} + O(x^{-2}).$$

The pole has exact order three only if the displayed coefficient is nonzero.

Theorem 9.3 (pure parity-fixed inverse-channel normal form). *For $m \geq 1$, $B > 0$, and $A_\alpha > 0$,*

$$ds^2 = Bx^2 dx^2 + x^{-2} \sum_{\alpha=1}^m A_\alpha dy_\alpha^2$$

has scalar curvature

$$R = -\frac{m(m+5)}{B} x^{-4}.$$

The coefficient is independent of all transverse amplitudes.

Proof. Set $\rho = \frac{1}{2}\sqrt{B}x^2$. The metric becomes a warped product $d\rho^2 + w(\rho)^2 h_{\text{flat}}$ with $w \propto \rho^{-1/2}$. The flat-fibre warped-product formula gives

$$R = -2m \frac{w''}{w} - m(m-1) \left(\frac{w'}{w} \right)^2 = -\frac{m(m+5)}{4\rho^2} = -\frac{m(m+5)}{Bx^4}.$$

□

Corollary 9.4 (asymptotic fixed face). *Assume a diagonal, even, C^2 polyhomogeneous germ for which*

$$g_{xx} = B(y)x^2 + O(x^4), \quad g_{\alpha\alpha} = A_\alpha(y)x^{-2} + O(1),$$

with positive B, A_α and uniform control of the stated remainders and transverse derivatives. Then

$$R = -\frac{m(m+5)}{B(y)} x^{-4} + O(x^{-2}).$$

STOP-PROOF. *Write the full asymptotic curvature expansion proving that transverse derivatives do not create another x^{-4} contribution. If the needed uniform/product hypotheses are stronger than stated, add them.*

9.1 Corners

The current general codimension-two statement is retained only in a monomial/polyhomogeneous form.

Proposition 9.5 (conditional corner vertex rule). **CONDITIONAL.** *Suppose a diagonal metric has specified monomial weights*

$$g_i = h_i(s) x^{p_i} y^{q_i} u_i(x, y, s),$$

where each u_i is a nonvanishing polyhomogeneous unit and the polar coefficients below are nonzero. Then the polar part of scalar curvature is supported on the candidate vertices

$$V_1 = (-(p_1 + 2), -q_1), \quad V_2 = (-p_2, -(q_2 + 2)), \quad V_0 = (-p_0, -q_0).$$

Along $x = \rho \epsilon^{a_x}$, $y = \epsilon^{a_y}$, the pole order is the support function of the surviving vertices.

STOP-PROOF. *Single-face pole orders and residue weights alone do not determine all mixed corner monomials. Include the closed symbolic identity for the coefficients of V_0, V_1, V_2 , derive it from the diagonal scalar-curvature formula, and state every cancellation condition. Do not extrapolate to codimension at least three.*

STOP-PROOF. *Deck parity $m \mapsto -m$, spin reflection $J \mapsto -J$, charge reflection $Q \mapsto -Q$, and defining-function gauge are distinct actions. Their common C_2 representation pattern is a bridge, not an identification.*

10 Kerr–Newman complementarity as boundary realization

Theorem 10.1 (open-stratum inverse-channel complementarity). *CONDITIONAL. For the selected metric $g_F(0)$:*

- (i) *scalar curvature is analytic in the open outer wedge, including on the Davies surface;*
- (ii) *on the open generic extremal stratum $T = 0$, away from its intersections with reflection faces, curvature has exact pole order three;*
- (iii) *on the open reflection-fixed strata $\Omega = 0$ and $\Phi_e = 0$, away from further intersections, curvature has exact pole order four and leading coefficient $-14/B$;*
- (iv) *the verified Schwarzschild intersection $\Omega = \Phi_e = 0$ has a direction-dependent Newton wedge.*

Dependency proof. Interior regularity is Corollary 6.2. The extremal germ is of the form in Proposition 9.2; its separately certified nonzero coefficient gives exact order three. Each open reflection face is the $m = 2$ case of Theorem 9.3. The Schwarzschild calculation is the specialized instance of Proposition 9.5. $\square$

STOP-PROOF. *Re-run and archive the factorization, interior-regularity, extremal noncancellation/Sturm, reflection-residue, and corner certificates. The intersections $T \cap \Omega$ and $T \cap \Phi_e$ are not classified and must be excluded or separately proved.*

STOP-PROOF. *For a path $Q = \epsilon$, $J = \rho\epsilon^a$ to approach the Schwarzschild corner one needs $a > 0$. The endpoint $a = 0$ approaches only the charge face with nonzero J and must not be described as a corner path.*

Remark 10.2 (scope). The 3/4/4 law is portable only conditionally: another diagonal inverse-channel surface must first be proved to have the corresponding local germs and nonzero coefficients. It is not a theorem about every framed constraint hypersurface.

11 Global continuation and arithmetic periods remain companions

11.1 Self-glue monodromy

For a self-glued algebraic relation, solving different output coordinates produces different branched covers. There is generally no single Galois or monodromy group attached to the unlabelled surface independently of the extraction coordinate.

Proposition 11.1 (local and global symmetry are independent). *BRIDGE. After lifting local chart-labelled data to every analytically continued solution sheet, global monodromy acts on the sheet index while local chart permutation acts on the coordinate index. When analytic continuation preserves the coordinate labels, the two actions commute; they are not canonically identified.*

STOP-PROOF. *The companion self-glue draft needs substantial repair before its general wreath laws are imported. In particular, $C_m \wr C_m$ is usually nonabelian and is not cyclic; the general gcd-layer kernel needs a Kummer or branch-cycle proof; a full directed-wreath realization requires independent within-block discriminant loci, not merely distinct block labels; and all quantifiers must respect the stated exceptional divisors.*

11.2 Corrected curvature constants

The elliptic curvature constants concern global integration cycles, third-kind differentials, an isogeny, and principal-value residue jumps. They are not used to prove the local orbit, metric-selection, or axial-cover theorems.

Theorem 11.2 (primary surface-to-link close-off). ESTABLISHED. *For*

$$S_s = \{D^2 + sDS + P^2 = 3\}, \quad s \neq 0,$$

let Γ_s^+ be either component of the spherical link of the asymptotic cone. Then

$$\int_{S_s} K_G dA = -2L(\Gamma_s^+).$$

At $s = 1$, exact Legendre reduction of $L(\Gamma_1^+)$ gives

$$\int_{S_1} K_G dA = -2^{7/4} \left[(3 + 2\sqrt{2})\Pi\left(\frac{4 - 3\sqrt{2}}{8}; \frac{2 - \sqrt{2}}{4}\right) - (2 + 2\sqrt{2})K\left(\frac{2 - \sqrt{2}}{4}\right) \right] = I_{\text{primary}}.$$

Proof. Orthogonally diagonalize the surface to $ax^2 - by^2 + z^2 = 3$, where $a = (1 + \sqrt{1 + s^2})/2$ and $b = (\sqrt{1 + s^2} - 1)/2$. On this cylinder the one-form

$$\eta = -\frac{2aby}{\sqrt{q}(a^2x^2 + z^2)}(x dz - z dx), \quad q = \|\nabla F\|^2,$$

is globally regular and satisfies $d\eta = K_G dA$. In the global hyperbolic parametrization its coefficient tends at the two oriented ends to

$$-\frac{\sqrt{ab}}{p(t)\sqrt{p(t) + b}}, \quad p(t) = a \cos^2 t + \sin^2 t.$$

The reciprocal tangent substitution

$$\tan t = \sqrt{\frac{a+b}{1+b}} \cot \theta$$

identifies the integral of this density with the length of one component of the spherical asymptotic link. Stokes' theorem therefore gives one copy from each end and proves the displayed factor -2 . The final formula is the established link-to-Legendre reduction. $\square$

STOP-PROOF. *Theorem 11.2 closes the primary scalar surface-to-period map. It does not identify the Galois-conjugate principal-value constant with a real region or relative cycle of the original DBP surface. A chain-level pushforward of the original curvature two-form to the elliptic differential would refine the primary statement, but is no longer needed to establish its scalar value; the dual surface cycle remains genuinely open.*

Remark 11.3. The primary and dual channels now have different geometric status. The primary period is the proved total curvature of S_1 . For the explicitly defined dual third-kind differential, the weighted residue is four and the two oriented indentations differ from the principal value by $\pm 4\pi i$, hence from one another by $8\pi i$; its interpretation in the original DBP realization still requires a new differential-and-cycle theorem.

12 What the paper absorbs and what it retires

The following results are internal to the main spine:

1. passive versus active output-chart correction;
2. the Active Recharting Orbit Theorem;
3. bordered-minor channel reduction and gauge transport;
4. the three-channel Gaussian specialization;
5. the family-relative Kerr–Newman metric selection;
6. generic and parity-fixed local curvature normal forms;
7. the generic normalization–reflection chain;
8. the axial definite-pencil, sheet-saturation, and least-root selection theorem.

The former curvature-orbit correction memo and canonical reduction note can be retired into the present text. The implementation-focused three-channel specification becomes a verification appendix. Detailed self-glue monodromy, elliptic periods, rotating horizon reconstruction, and full arithmetic certificates remain companion work.

13 The remaining unification theorem

Conjecture 13.1 (intrinsic orbit–quotient–selection principle). *There is a class of systems equipped with*

(carrier, symmetry, quotient, real chamber, selected section, wall data)

and morphisms preserving these data, such that:

- (i) the local active-recharting systems and global horizon-cover systems are objects of the class;*
- (ii) invariant observables factor through the quotient;*
- (iii) physical observables require the selected section;*
- (iv) selected sections avoid the interior branch locus;*
- (v) boundary degenerations are typed functorially by stabiliser and wall data.*

STOP–PROOF. *This is not a theorem until the data are minimized, native morphisms and equivalence are defined, closure is proved, and a third independent realization is exhibited. It must not be used to prove any preceding result.*

14 Conclusion

The unified framework is a local-to-global theorem chain. The active chart-jet orbit is the local carrier; elementary shape curvature is a reduced projection; channel data is a framed account. A physical realization then requires selection: in Kerr–Newman geometry, positivity and interior regularity select a metric-family member; in the axial cover, definiteness and orientation select the first boundary root. Involutions remove odd data and organize fixed strata, while local metric germs determine boundary valuations.

What has been unified is the logical architecture and dependency of proved results. What has not been proved is a categorical equivalence of the realizations. Keeping that boundary explicit is what makes the synthesis auditable.

A Source absorption map

Source manuscript	Disposition	Status
Active role-jet orbit calculus	Foundational local theorem, with faithfulness correction	Absorbed
Canonical Invariant Reduction and Gauge Transport	Channel polynomial and gauge section	Absorbed
Theorem 8.1 curvature correction	Passive/active distinction and value-plus-jet guardrail	Retired memo
Three-channel K_G specifications	Gaussian specialization and verification appendix	Absorbed in part
Kerr–Newman inverse-channel metric	Selection and complementarity realization	Conditional
Parity-fixed normal forms	Local valuation section	Absorbed; corner proof owed
Galois horizon cover and R5–R7 entries	Global cover module	Conditional arithmetic
Self-glue monodromy	Global continuation companion	Not load-bearing
Corrected curvature constants	Primary surface-to-link theorem absorbed; dual arithmetic-period companion	Primary established; dual geometric cycle open

B Notation collision ledger

Concept	Unified notation	Avoid
mass square	$U = m^2$	metric-family u
coordinate–coordinate metric weight	η	mass-square u
channel polynomial variable	v	mass-square U
gradient norm	$q = g^\top g$	charge Q without context
active chart group	$G_{\text{act}} = S_3$	global monodromy group
sign group	$H_{\text{sign}} = (C_2)^k$	axial deck group
axial deck group	$G_{\text{deck}} = \langle \tau \rangle$	charge reflection
mass chamber	$\mathcal{C}$	outer KN wedge W_+
channel tuple	$(\kappa_c, \kappa_{\text{int}}, \kappa_s)$	alternating tuple orders

C Bibliographic placeholders

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